Electric Horizons: Mapping the Future of Clean Mobility and Air

Final Project Proposal
FA550A/WS: Data Visualization Applications

Achintya
CWID: 20021326

Title: *Driving Clean: Visualizing EV Adoption, Infrastructure, and Environmental Impact*

Introduction

The shift to electric vehicles (EVs) represents not only a technological revolution but a global environmental imperative. As nations confront the dual challenge of reducing greenhouse gas emissions and enhancing urban air quality, EV adoption has emerged as a critical tool in sustainable transportation policy. Simultaneously, air quality has become a public health priority, with pollutants such as PM2.5, NO_2, and CO contributing to respiratory and cardiovascular diseases worldwide. This project aims to explore how the increasing adoption of EVs correlates with improvements in air quality and reductions in CO_2 emissions.

In addition, the availability and growth of EV infrastructure, including charging stations and supportive policy incentives, plays a central role in accelerating this transition. Another key consideration is the sustainability and equity of this transformation, which hinges on access to critical resources such as lithium and cobalt. By analyzing these interconnected domains, this project will provide a multi-dimensional perspective on EV adoption and its environmental footprint using interactive, accessible dashboards built in Tableau, supported by analysis and preprocessing in R and Python.

The broader goal of this project is to inform policy, increase awareness, and visualize the progress and challenges of the electrification movement. Through rich datasets and meaningful visualizations, this project will offer insights into which regions are leading or lagging, what environmental gains are measurable, and what infrastructural or systemic gaps remain.

Project Objectives

1. Track global and regional trends in EV adoption by powertrain type.

2. Explore the correlation between EV growth and improvements in air quality metrics (PM2.5, NO_2, CO).
3. Analyze the relationship between EV adoption and CO_2 emissions per capita across countries.
4. Visualize and compare the development of EV charging infrastructure and its impact on sales growth.
5. Identify global clusters of countries based on EV infrastructure availability, emissions, and adoption.
6. Evaluate seasonal trends in air pollution and EV market share.
7. Develop forecasts and predictive models for future EV adoption and air quality trends.
8. Create an interactive Tableau story with geospatial, time-series, and categorical visualizations that reflect the above themes.

Research Questions

We propose some of the following research questions, which will guide our visual and statistical analyses:

🔌 **EV Adoption Trends**

1. How has global EV adoption evolved over the past decade, and what are the regional variations?
2. Which regions are leaders or laggards in the adoption of battery-electric vehicles
3. How does the availability of EV charging- infrastructure influence EV sales and adoption trends?

🌍 **Air Quality & Emissions Impact**

4. Is there a measurable correlation between EV adoption and improvements in air quality (e.g., PM2.5, NO_2)?
5. What is the relationship between EV growth and CO_2 emissions per capita across countries?
6. Are there seasonal or temporal trends in air pollution that correspond with increased EV market share?

Visualization Tools and Justification

- **Tableau:** Will be used to build interactive dashboards that allow exploration of temporal, geographic, and categorical trends. Ideal for multi-layered visual storytelling.
- **R :** Used for initial data wrangling, statistical modeling (correlation, regression), and customized static visualizations. R is powerful for time-series and cluster analysis preparation before importing visuals into Tableau.
- **Python (**Jupyter, pandas, numpy, seaborn**):** Used for initial EDA, data cleaning, and transformation.

Tools and Methods

- **Python (Jupyter):**
  Used for data preprocessing, transformation, and exploratory analysis. Libraries:
  - pandas for cleaning and manipulation
  - numpy for numeric computations
  - matplotlib/seaborn for base plots
  - scikit-learn for clustering and predictive modeling
- Time-Series Analysis: Trends in EV growth, pollutant levels, CO_2 emissions
- Correlation Modeling: Adoption vs. air quality / CO_2 emissions
- Clustering: Segmenting countries based on adoption, infrastructure, and pollution metrics
- Geospatial Mapping: EV infrastructure and AQI levels
- Treemaps/Bar Charts: Powertrain types, emissions, EVSE availability

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Data Sources:

(Additional Data sets other than these might be used)

Dataset	**Description**	**Link**
IEA EV Data	EV sales, market share, BEV/PHEV breakdown	IEA EV Data

OpenAQ / WHO Air Quality	PM2.5, NO2, CO levels by region and year	OpenAQ API
Our World In Data - CO_2 Emissions	Per capita emissions by country	OWID CO2
EV Charging Infrastructure (IEA, Statista)	Global EVSE stations by region	IEA Infrastructure Report
US Department of Energy / UN Resource Data	Lithium/cobalt availability and extraction	DOE

Data Summary

I have chosen multiple datasets that are both globally recognized and rich in spatial, temporal, and categorical detail. These datasets are particularly valuable because they capture both quantitative and contextual dimensions of EV adoption and environmental impact.

I am particularly interested in leveraging:

- **IEA EV Dataset**: This dataset includes EV sales, powertrain type (BEV, PHEV), and market share from 2014 to 2023. It allows comparison of regional adoption trends and segmentation by vehicle technology.
- **OpenAQ / WHO Air Quality Dataset:** Includes pollutant-specific indices such as PM2.5, NO_2, and CO levels across cities and time. This dataset is vital for understanding environmental benefits tied to EV penetration.
- **OWID CO_2 Emissions**: Offers longitudinal emissions data by country. It helps contextualize emission reduction claims in relation to EV rollout.
- **EV Infrastructure Data (IEA, Statista):** Tracks public and private EV charging stations by country. This dataset is key to understanding the infrastructural factors that support or hinder EV adoption.
- **DOE / UN Resource Data:** Information on global lithium, cobalt, and other rare earth minerals. This enables analysis of supply-side risks and resource constraints in EV growth.

Together, these datasets support multi-level analysis including correlation, forecasting, geospatial mapping, and clustering. They align well with the project goals of visualizing and analyzing EV-environmental interactions over time and geography.

Methodology and Analyses

- **Time-Series Analysis:** EV sales, charging stations, air quality metrics over time
- **Correlation Analysis:** Between EV adoption and pollutant levels or CO_2 emissions
- **Cluster Analysis:** Countries grouped by adoption, infrastructure, emissions
- **Forecasting:** Projecting future EV growth and air quality improvements
- **Geospatial Analysis:** Maps of EV infrastructure density vs. air quality
- **Treemaps/Bar Charts:** Breakdown of EV types and emission sources

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Proposed Project Timeline

Date	Milestone
4/12	Proposal ready for submission- objectives, datasets, research questions, and methods
4/16	Finalize datasets, clean and merge air quality + EV data
4/20	Initial EDA and statistical exploration completed
4/24	Narrative addressing research questions drafted; presentation structure begins
4/27	Progress review - Narrative + Core visualizations
5/1	Complete correlation/cluster analysis and infrastructure impact models
5/5	Final dashboard and Tableau story polish
5/10	Complete presentation deck and narrative script
5/11	Final submission with dashboards, insights, and documentation

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Planned Sample Visualizations :

I will be using the following visualization types to support my final Tableau project. Each is selected to answer specific research questions and uncover meaningful patterns within the data:

1. **Choropleth Map:** EV adoption by country (BEV vs. PHEV)
2. **Time-Series:** AQI vs. EV growth (yearly and regionally)
3. **Treemap:** Emissions sources by country (CO_2, PM2.5, NO_2)
4. **Bar Chart:** Top 10 countries by EV infrastructure vs. adoption

5. **Scatter Plot + Trend Line:** EV adoption vs. AQI improvement
6. **Cluster Diagram:** Countries grouped by growth rate, emissions, infrastructure
7. **Line + Area Chart:** Charging station growth alongside sales growth

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Conclusion

This project aspires to provide a comprehensive, data-driven narrative about the evolution and impact of EV adoption worldwide. By blending air quality, EV infrastructure, emissions, and global market data, we aim to produce dashboards that are not just informative but actionable—helping policymakers, industry stakeholders, and citizens visualize what the future of clean transportation might look like.